

10.18

a、

父親 12.15%，母親 11.68%。

b、

	educ	MOTHERCOLL	FATHERCOLL
educ	1.0000000	0.3594705	0.3984962
MOTHERCOLL	0.3594705	1.0000000	0.3545709
FATHERCOLL	0.3984962	0.3545709	1.0000000

教育對於工資的影響可能不是線性的，相關係數又顯示了IV的相關性

c、

```
Call:
lm(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
  exper + I(exper^2), data = mroz_work)

Residuals:
    Min       1Q   Median       3Q      Max
-3.08719 -0.32444  0.04147  0.36634  2.35621

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.132761  0.4565325  -0.289 0.7744
educ         0.0760180  0.0394077  1.929 0.0549
exper        0.0433444  0.0134135  3.231 0.00153 **
I(exper^2)   -0.0008711  0.0004017  -2.169 0.03066 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-squared: 0.147, Adjusted R-squared: 0.1409
Wald test: 8.2 on 3 and 424 DF, p-value: 2.569e-05

2.5 %    97.5 %
educ -0.001219763 0.1532357
```

95% 以下不顯著

d、

```
Call:
lm(formula = educ ~ MOTHERCOLL + exper + I(exper^2), data = mroz_work)

Residuals:
    Min       1Q   Median       3Q      Max
-7.4267 -0.4826 -0.3731  1.0000  4.9353

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.079094  0.303118  39.849 < 2e-16 ***
MOTHERCOLL  2.517068  0.315713  7.973 1.46e-14 ***
exper        0.056230  0.042101  1.336  0.182
I(exper^2)   -0.001956  0.001256  -1.557  0.120
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.133 on 424 degrees of freedom
Multiple R-squared: 0.1347, Adjusted R-squared: 0.1285
F-statistic: 21.99 on 3 and 424 DF, p-value: 2.965e-13
```

Linear hypothesis test:

MOTHERCOLL = 0

Model 1: restricted model

Model 2: educ ~ MOTHERCOLL + exper + I(exper^2)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	424	1929.9	1	289.32	63.563	1.455e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

21.99 > 10, MOTHERCOLL 是一個好 IV，與教育程度相關

e、

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
  FATHERCOLL + exper + I(exper^2), data = mroz_work)

Residuals:
    Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819  0.3922213  -0.712 0.47714
educ         0.0878477  0.0307808  2.854 0.00453 **
exper        0.0426761  0.0132950  3.210 0.00143 **
I(exper^2)   -0.0008486  0.0003976  -2.135 0.03337 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-squared: 0.153, Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06

2.5 %    97.5 %
educ 0.02751845 0.1481769
```

問題比 c 中，增加有效的 IV 能提高估計的精確度

1x educ 變顯著

f、

```
Call:
lm(formula = educ ~ MOTHERCOLL + FATHERCOLL + exper + I(exper^2),
  data = mroz_work)

Residuals:
    Min       1Q   Median       3Q      Max
-7.2152 -0.3056 -0.2152  0.7627  5.0620

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 11.890859  0.290251  40.965 < 2e-16 ***
MOTHERCOLL  1.749947  0.322347  5.429 9.58e-08 ***
FATHERCOLL  2.186612  0.329917  6.628 1.04e-10 ***
exper        0.049149  0.040133  1.225  0.221
I(exper^2)   -0.001449  0.001199  -1.209  0.227
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.033 on 423 degrees of freedom
Multiple R-squared: 0.2161, Adjusted R-squared: 0.2086
F-statistic: 29.15 on 4 and 423 DF, p-value: < 2.2e-16
```

Linear hypothesis test:

MOTHERCOLL = 0

FATHERCOLL = 0

Model 1: restricted model

Model 2: educ ~ MOTHERCOLL + FATHERCOLL + exper + I(exper^2)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	423	1748.3	2	470.8	56.963	2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Wu-Hausman test $p=0.492$ ，無法拒絕 educ 為外生的假設

Sargan test $p=0.620$ IV 滿足外生性條件，無過度識別的問題

g、

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
  FATHERCOLL + exper + I(exper^2), data = mroz_work)

Residuals:
    Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819  0.3922213  -0.712 0.47714
educ         0.0878477  0.0307808  2.854 0.00453 **
exper        0.0426761  0.0132950  3.210 0.00143 **
I(exper^2)   -0.0008486  0.0003976  -2.135 0.03337 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-squared: 0.153, Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06

Diagnostic tests:
df1 df2 statistic p-value
Weak instruments 2 423 56.963 <2e-16 ***
Wu-Hausman      1 423 0.519 0.472
Sargan          1 NA 0.238 0.626
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2個IV是有效性的，模型設定是正確的。

10.20

a、

```
Call:
lm(formula = msft_ex ~ mkt_ex, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27424 -0.04744 -0.00820  0.03869  0.35801

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003250   0.006036   0.538   0.591
mkt_ex       1.201840   0.122152   9.839 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523,    Adjusted R-squared:  0.3486
F-statistic: 96.8 on 1 and 178 DF,  p-value: < 2.2e-16

OLS 估計的 Beta 值為: 1.20184
```

beta > 1, risky.

b、

```
Call:
lm(formula = mkt_ex ~ RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.110497 -0.006308  0.001497  0.009433  0.029513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.983e-02  2.195e-03  -36.0 <2e-16 ***
RANK         9.067e-04  2.104e-05   43.1 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126,    Adjusted R-squared:  0.9121
F-statistic: 1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

RANK 變 market risk premium 高度相關, 3% IV

c、

```
Call:
lm(formula = msft_ex ~ mkt_ex + vhat_b, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.005904   0.504   0.6146
mkt_ex       1.278318   0.126749  10.085 <2e-16 ***
vhat_b       -0.874599   0.428626  -2.040  0.0428
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,    Adjusted R-squared:  0.36
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

vhat 在 95% CI 下不顯著 + market risk premium 存在内生性問題
91% CI 下不顯著

d、

```
Call:
ivreg(formula = msft_ex ~ mkt_ex | RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.271625 -0.049675 -0.009693  0.037683  0.355579

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.006044   0.499   0.618
mkt_ex       1.278318   0.128011   9.986 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-Squared:  0.3508,    Adjusted R-squared:  0.3472
Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16
```

↳ 6
↳ Attenuation Bias

e、

```
Call:
lm(formula = mkt_ex ~ RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.109182 -0.006732  0.002858  0.008936  0.026652

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0804216  0.0022622  -35.55 <2e-16 ***
RANK         0.0009819  0.0000400   24.55 <2e-16 ***
POS         -0.0092762  0.0042156   -2.20  0.0291 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149,    Adjusted R-squared:  0.9139
F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16
```

Linear hypothesis test:

RANK = 0
POS = 0

Model 1: restricted model
Model 2: mkt_ex ~ RANK + POS

	Res.DF	RSS	DF	Sum of Sq	F	Pr(>F)
1	179	0.43784				
2	177	0.03727	2	0.40057	951.26	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

↳ -pos 導致 R² 上升
(RANK, POS 為好的 IV 組合)

f、

```
Call:
lm(formula = msft_ex ~ mkt_ex + vhat_e, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27132 -0.04261 -0.00812  0.03343  0.34867

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.005972   0.503   0.6157
mkt_ex       1.283118   0.126344  10.156 <2e-16 ***
vhat_e       -0.954918   0.433062  -2.205  0.0287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,    Adjusted R-squared:  0.3625
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

(% 顯著水準下, 無法行指出 market risk premium 為内生 (0.0287 > 0.01))

g、

```
Call:
ivreg(formula = msft_ex ~ mkt_ex | RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27168 -0.04960 -0.00983  0.03762  0.35543

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.006044   0.497   0.62
mkt_ex       1.283118   0.127866  10.035 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared:  0.3507,    Adjusted R-squared:  0.347
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16
```

IV 估計的 beta 更能反映 mkt
變市場的相關係